Descriptive Analytics Report:

Order Volume, Seller & Regional Demand at Scale

Tracking Best-Selling Products, Ordering Trends, Ratings, Regional Demand, Payment Behavior & Seller/Inventory Health

Meesho — Simulated E-Commerce Marketplace Dataset
Report window: 1 month | 40 Products | 8 categories | 10 states

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1. Note on Data & Objective

Meesho's live order and seller data isn't publicly accessible, so this report uses a synthetic dataset built to mirror real Indian social-commerce marketplace patterns (40 products across 8 categories, 10 states, 1 month, ~12.5 million simulated orders). It is designed to show exactly what a descriptive analytics pipeline processing millions of daily order events would produce — the same structure applies once real data is substituted in.

Objective

Build a descriptive analytics layer over daily ordering and seller activity to answer seven recurring business questions:

- What is being ordered most right now, and by how many customers?
- How is total order volume and GMV trending day over day?
- How are products rated, and where is rating quality slipping?
- Where is order demand concentrated geographically, and by category?
- How does customer behavior — payment mode, return rate — vary by region and category?
- Which sellers and seller tiers are driving the most platform revenue?
- Which products are at risk of stocking out before demand fades?

2. How This Would Work at Real Scale

This report was actually produced with a lightweight, practical toolchain — the same one a solo analyst or a small seller-ops team would reach for before any big-data infrastructure is justified, built entirely around VS Code as the daily driver:

Layer	Purpose	Typical Tools
Data source	Raw order/seller records, exported on a schedule	CSV / API export from the source system
Storage & querying	Load the export into a queryable table, join/filter/aggregate	PostgreSQL, queried via VS Code's SQLTools extension or DBeaver
Analysis	Clean, reshape, and compute the metrics behind every chart in this report	Python (pandas, numpy) written and run in VS Code with the Python & Jupyter extensions
Visualization	Generate each figure from the aggregated tables	Python (matplotlib, seaborn) in VS Code
Version control	Track changes to scripts, queries, and report templates over time	Git, with VS Code's built-in Source Control panel and GitHub
Reporting	Assemble the narrative, tables, and charts into a shareable document	Scripted PDF report (Python + ReportLab), run from VS Code's integrated terminal

This stack scales further than it looks: PostgreSQL handles the aggregation once row counts get too large for a flat file alone, and the same pandas/matplotlib scripts — version-controlled in the same VS Code workspace — can be pointed at a warehouse table instead of a CSV without changing the analysis logic itself. The jump to a Kafka/Spark/Redshift-style pipeline only becomes necessary once ingestion needs to happen continuously, in real time, at millions of events per day — for a daily or weekly reporting cadence like this one, VS Code + PostgreSQL + Python is the right-sized tool, not an early version of a bigger system.

The report that follows reflects the output of that workflow — daily aggregated tables computed in pandas and charted with matplotlib — not raw event-level data, since no report renders millions of individual rows usefully.

3. Daily Ordering Trend

Over the simulated 1-month window: ~12.7 million orders logged, generating ~Rs. 536 Crore in GMV, with a weekly seasonal cycle (mid-week dips, weekend peaks) layered on top of gradual category-level growth or decline trends per product.

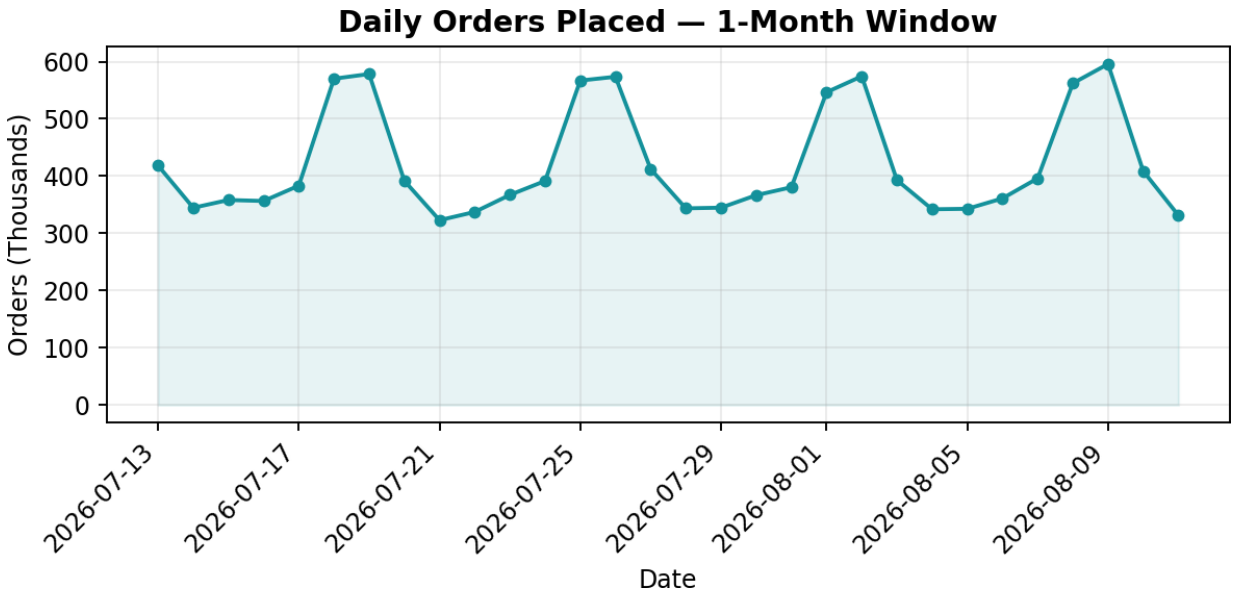


Figure 1. Daily orders placed across the 1-month window. Weekend peaks and mid-week dips form a repeating cycle.

Key Insight: Ordering is highly cyclical, not growing — volume oscillates roughly between 340K and 590K orders/day on a repeating weekly pattern with no clear upward or downward trend over the 1 month.

KPI Definitions Used

- **Daily orders** = count of confirmed checkout events across all sellers that day
- **Daily GMV** = sum of order value at checkout for that day, before returns/cancellations net out

3.1 Daily GMV Trend

GMV broadly tracks the order-volume curve but is not identical to it — days with heavier ordering in higher-AOV categories (Jewellery & Accessories, Home Furnishing) push GMV up even without a matching spike in order count.

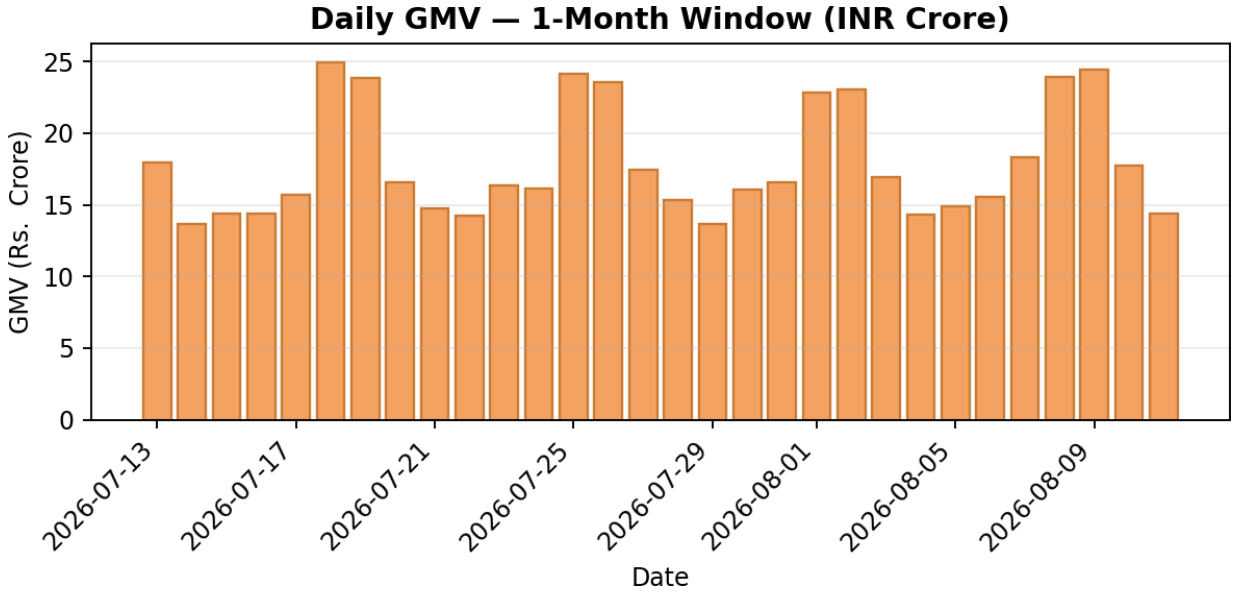


Figure 2. Daily GMV in Rs. Crore. Compare the shape against Figure 1 — divergence signals a shift in category mix.

Key Insight: GMV doesn't track order count 1:1 — some lower-volume days out-earn higher-volume ones, pointing to shifts in category mix (higher-AOV categories) or average basket size.

4. Best-Selling Products (Most Recent Day)

Top performer led with 22,299 units sold in a single day. This is a common "hero product" pattern on marketplaces: a small number of SKUs drive a disproportionate share of orders — a Pareto effect worth flagging for homepage merchandising and warehouse/fulfillment capacity planning.

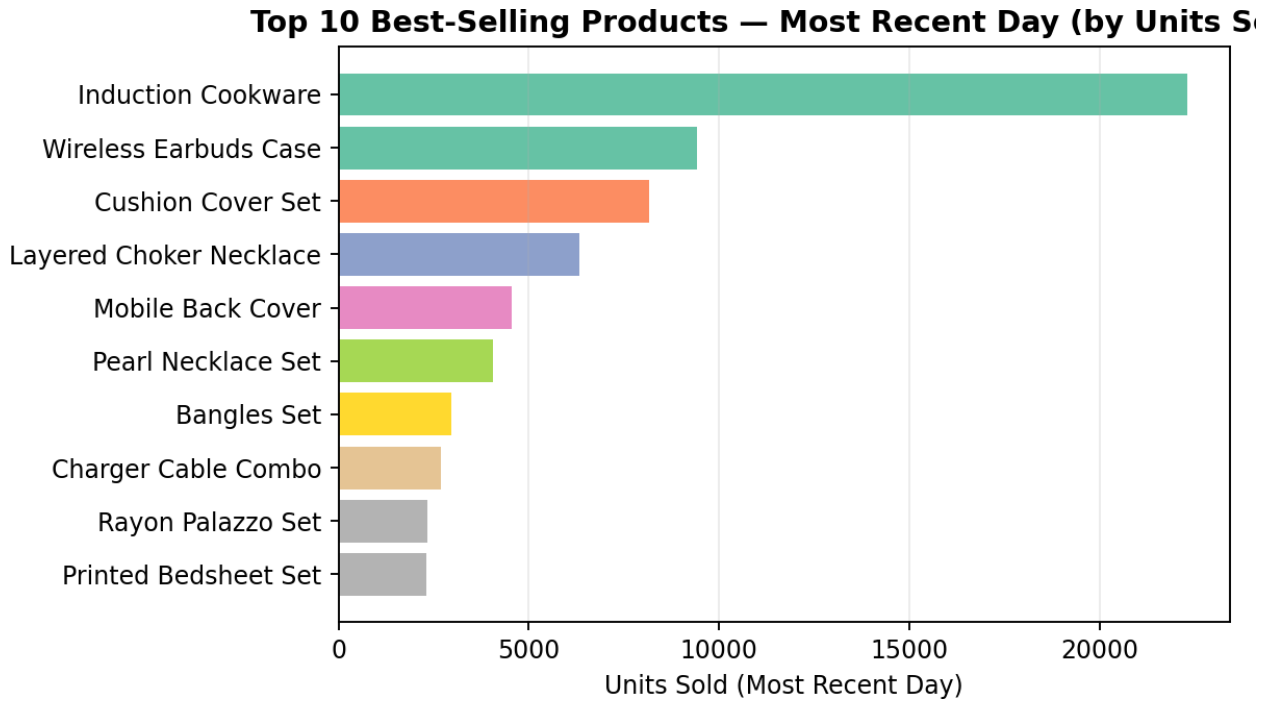


Figure 3. Top 10 products by units sold on the most recent day.

Key Insight: One hero product dominates — "Induction Cookware" (22,299 units) leads by a wide margin over #2, a classic Pareto pattern worth flagging for fulfillment capacity planning.

Ranking logic: units sold, not reach — a product with fewer buyers but a very high repeat-purchase rate can outrank a product with more, one-time buyers, so both cuts (units and reach) should exist as separate leaderboard views in a real dashboard.

4.1 Best-Selling Products by New Customer Reach

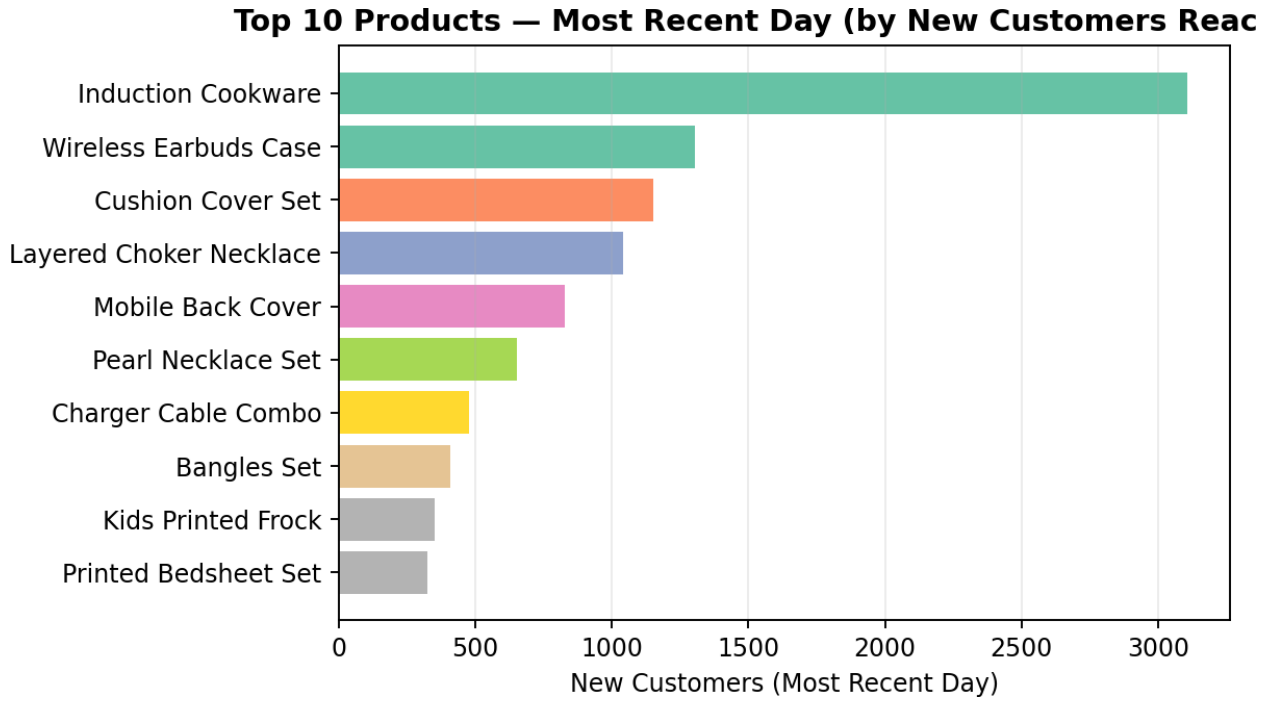


Figure 4. Top 10 products by new customers reached on the most recent day — a different leaderboard from Figure 3.

Key Insight: Reach and volume rankings diverge — some products rank high on new customers but not on the units-sold list, meaning some products pull in breadth (many new customers) while others pull depth (repeat orders from existing customers).

Product	Category	New Customers
Induction Cookware	Footwear	3106
Wireless Earbuds Case	Kids Fashion	1303
Cushion Cover Set	Jewellery & Accessories	1152
Layered Choker Necklace	Men Apparel	1041
Mobile Back Cover	Women Ethnic Wear	828
Pearl Necklace Set	Kitchen & Home	654

5. Rating Distribution

Average rating across all product-days: 3.93 / 5, clustered tightly (most mass between 3.5 and 4.4). In a real system, this view is most useful segmented, not aggregated.

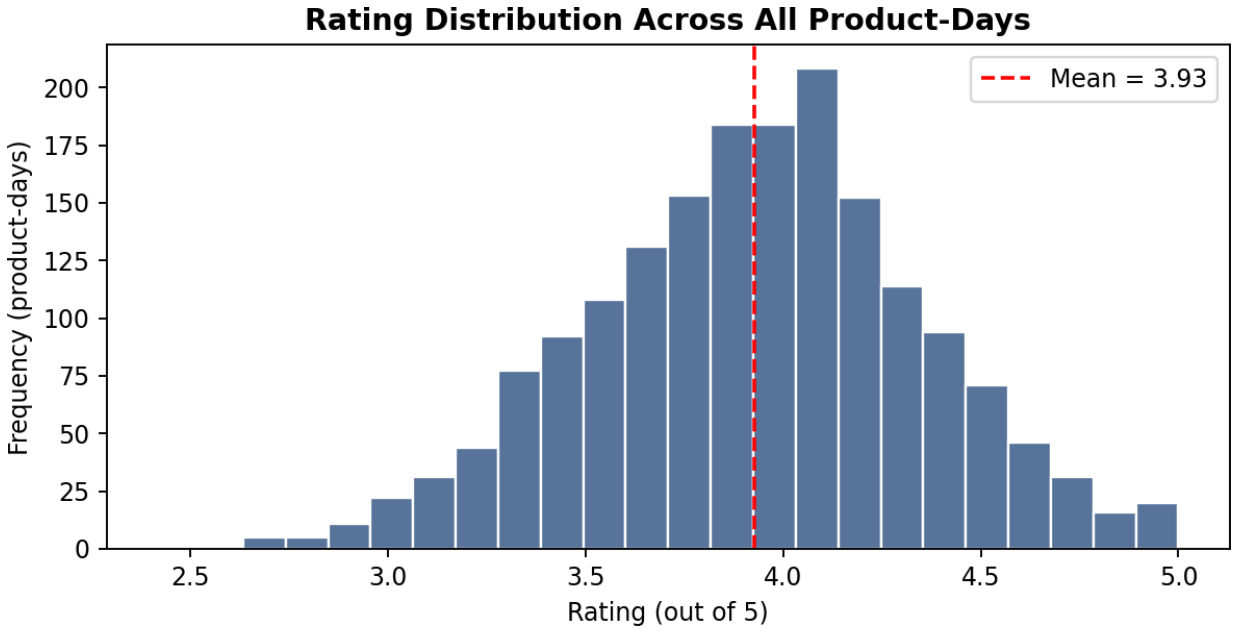


Figure 5. Distribution of ratings across all product-days in the window.

Key Insight: Ratings cluster tightly and skew positive — mean 3.93/5, most mass between 3.5–4.4, so a single average score hides meaningful outliers.

• **Products trending downward** in rolling 7-day average rating — the earliest signal of a quality, sizing, or fulfillment issue

• **Products with high order volume but sub-3.5 ratings** — the highest-risk combination, since volume amplifies the reach of poor reviews and returns

5.1 Products with Declining Rating Trend

6 products show a sustained downward drift in their 7-day rolling average rating over the window — the earliest, cheapest signal that something in product quality, sizing accuracy, or packaging has changed before ratings collapse at scale.

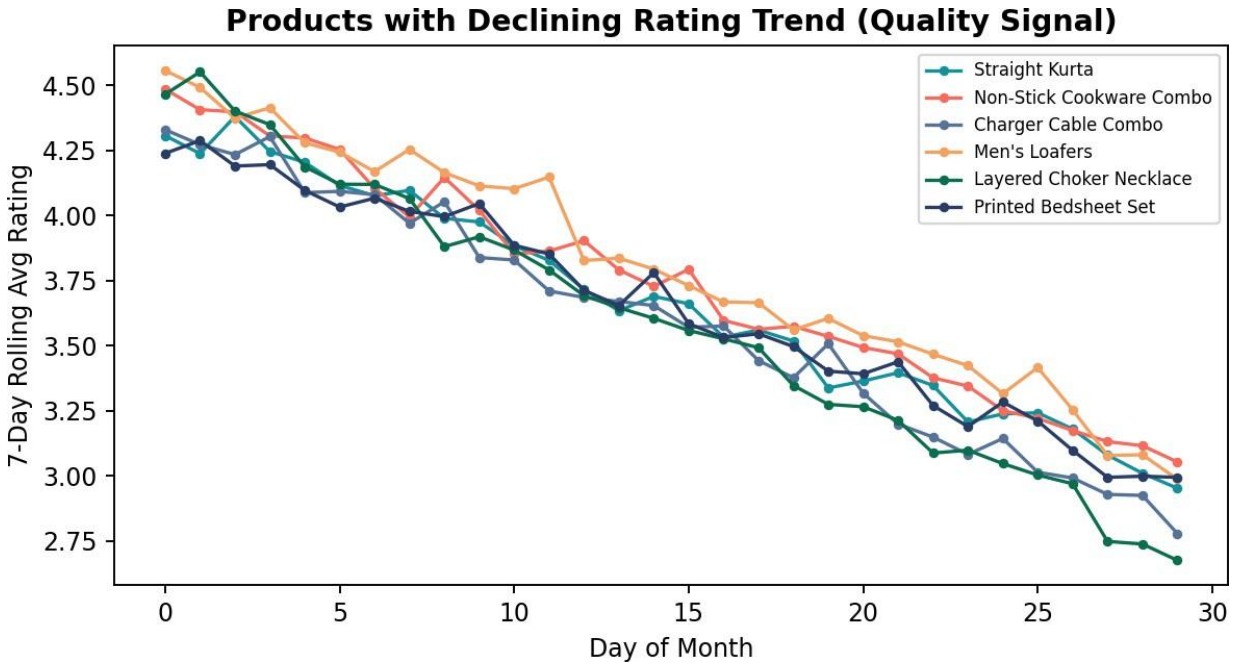


Figure 6. Rolling 7-day average rating for products flagged with a declining trend.

Key Insight: 6 products are bleeding quality signal — the steepest drops fall toward ~2.6–3.1, an early warning ahead of any ratings collapse.

Product ID	Product
MSH-2017	Straight Kurta
MSH-2037	Non-Stick Cookware Combo
MSH-2034	Charger Cable Combo
MSH-2018	Men's Loafers
MSH-2032	Layered Choker Necklace
MSH-2001	Printed Bedsheet Set

5.2 High-Risk Combination: High Order Volume, Sub-3.5 Rating

These products combine above-median monthly order volume with a current rating below 3.5 — the highest-risk combination on the platform, since every additional order reaches more customers who may then leave the same critical review or request a return.

Product	Monthly Orders	Rating
Foldable Travel Duffle	208,043	2.98
Kids Printed Frock	202,666	3.07
Mobile Back Cover	201,564	3.07
Plastic Storage Container	169,823	3.11
Boys Track Suit	165,366	3.14
Cushion Cover Set	122,820	3.21
Charger Cable Combo	96,665	3.46
Table Runner	79,800	3.55

Key Insight: "Foldable Travel Duffle" is the single biggest risk on the platform — 208,043 monthly orders at a 2.98 rating means poor reviews are reaching the widest possible audience.

Recommended action: route these products to a manual quality-check / seller-compliance review queue before the next homepage promotional push increases their exposure further.

6. Regional Demand

Order distribution across states was fairly balanced in this simulation (Uttar Pradesh and Maharashtra led). In practice, regional demand at Meesho's real scale typically shows heavier concentration in Tier-2/Tier-3 states, with metro-heavy states contributing disproportionately to GMV despite lower order counts.

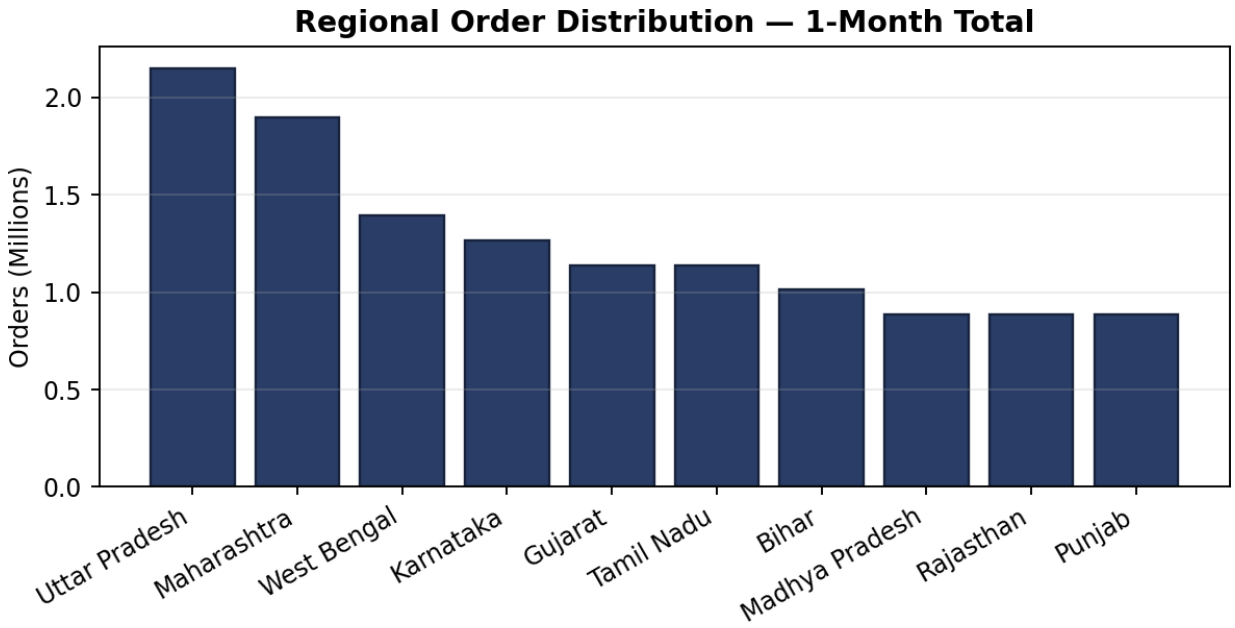


Figure 7. Orders by state over the 1-month window.

Key Insight: Uttar Pradesh alone drives the largest share of orders (2,152,314) — demand is concentrated, not evenly spread, which should anchor regional warehouse and last-mile investment priority.

This view is what drives:

- Regional warehouse and hub placement
- Localized language support and regional seller onboarding
- Last-mile courier capacity planning (demand spikes in states with limited local coverage)

6.1 Category-Wise Performance

Layering category onto the regional view surfaces where GMV concentration actually sits, independent of raw order-count numbers.

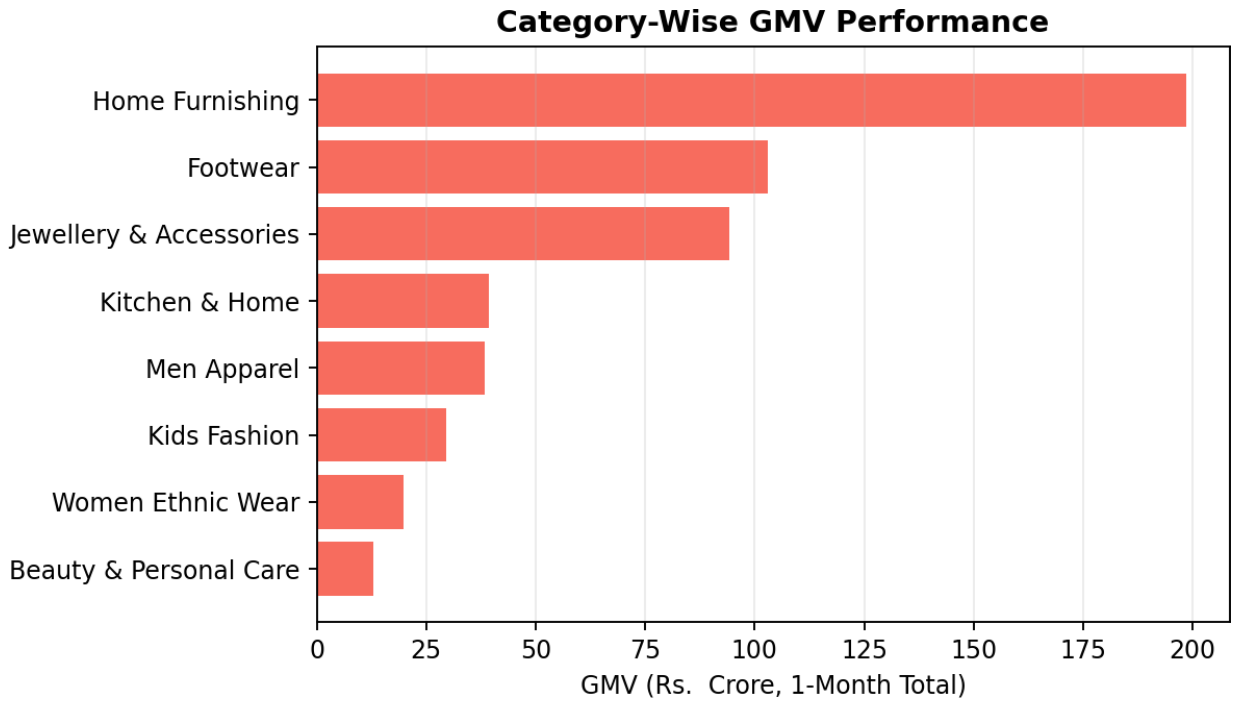


Figure 8. GMV by category, 1-month total, ranked highest to lowest.

Key Insight: Home Furnishing leads GMV but notably outperforms its order-count share — meaning category mix, not just order volume, drives revenue.

6.2 Category Share of Orders

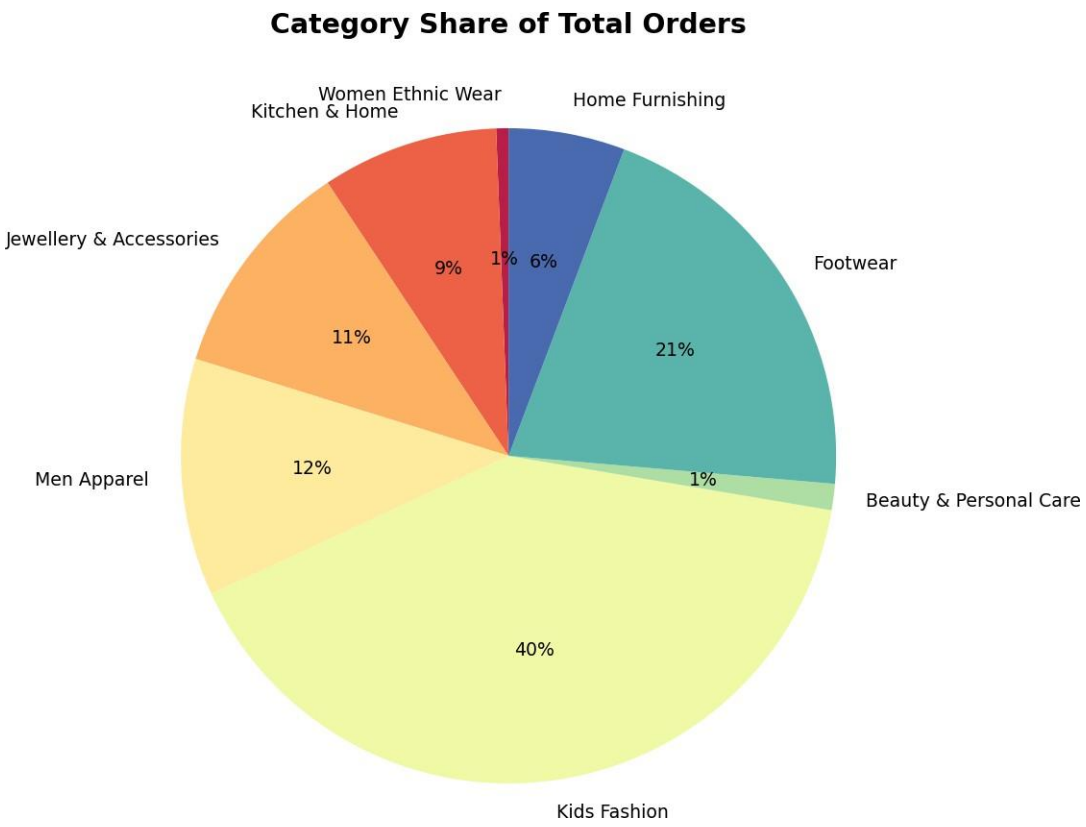


Figure 9. Share of total orders by category.

Key Insight: Kids Fashion and Footwear are the volume backbone, while Women Ethnic Wear remains marginal, suggesting a long-tail catalog propped up by a handful of core categories.

6.3 Day-of-Week Ordering Pattern

Aggregating the 1-month window by weekday isolates the seasonal cycle seen in Figure 1. Sunday is the strongest day for ordering, while Tuesday is consistently the softest — a pattern that should directly inform flash-sale scheduling, homepage merchandising, and push-notification timing.

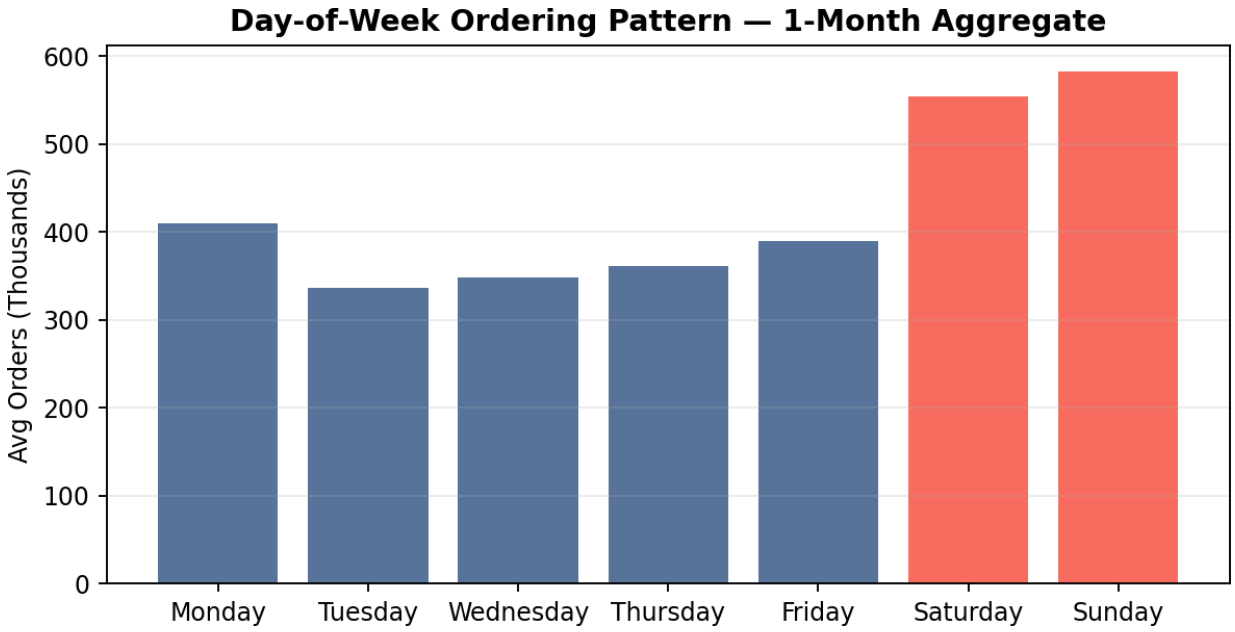


Figure 10. Average orders by day of week (weekend days highlighted).

Key Insight: Weekends convert to notably higher ordering — Saturday/Sunday clearly outperform the flat Mon–Fri baseline, with Tuesday the softest day — a strong signal for flash-sale timing.

6.4 Payment Mode Mix

Cash on Delivery leads order share at 47%, consistent with Meesho's Tier-2/Tier-3-heavy customer base. Payment mode matters operationally: COD orders carry different fraud, reverse-logistics, and working-capital implications than prepaid orders.

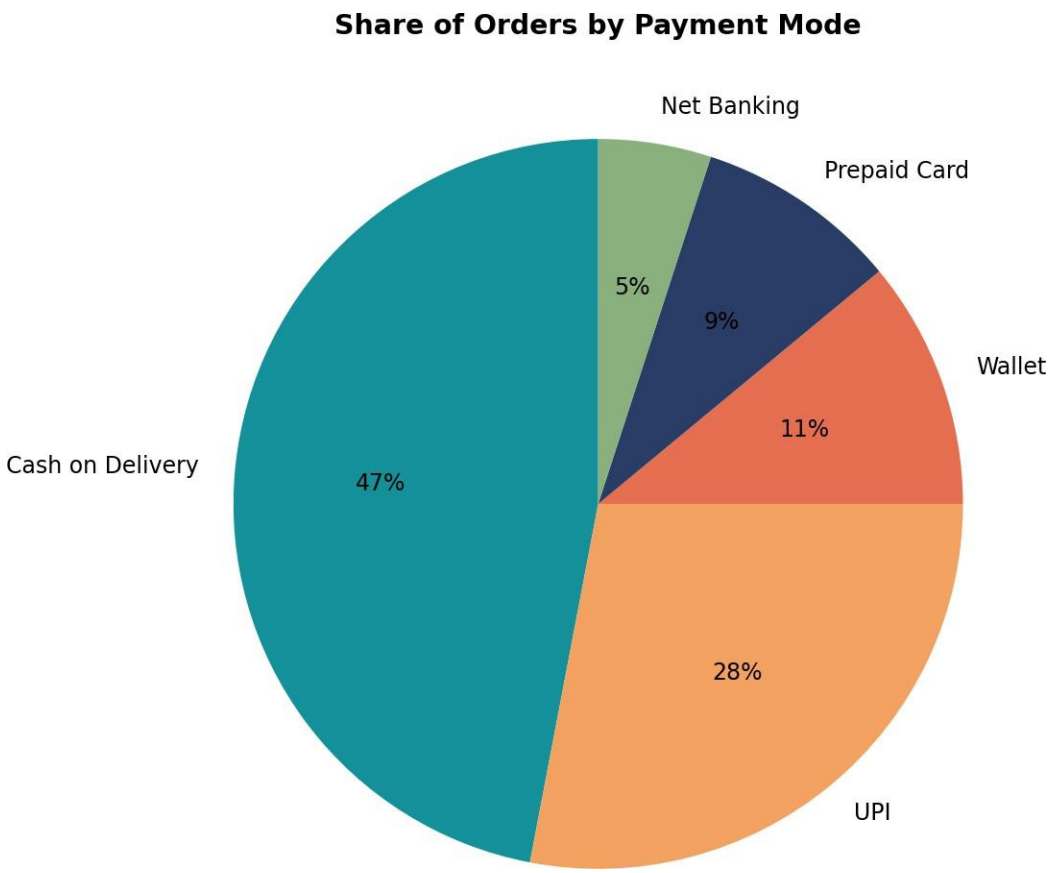


Figure 11. Share of orders by payment mode.

Key Insight: COD remains the dominant payment mode at 47%, more than UPI and Wallet combined trailing individually — confirms cash-on-delivery trust, not prepaid convenience, still drives the bulk of orders.

Payment Mode	Order Share	Orders
Cash on Delivery	47%	5,950,516
UPI	28%	3,544,988
Wallet	11%	1,392,674
Prepaid Card	9%	1,139,460
Net Banking	5%	633,033

6.5 Order Status & Return Rate by Category

Return rates vary meaningfully by category — Jewellery & Accessories runs highest at ~14.7% returned, typical for fit-sensitive apparel categories, while Home Furnishing sits lowest at ~6.5%, as expected for non-fit-dependent categories. Return and cancellation rates should be tracked as a leading indicator alongside the rating metrics in Section 5.

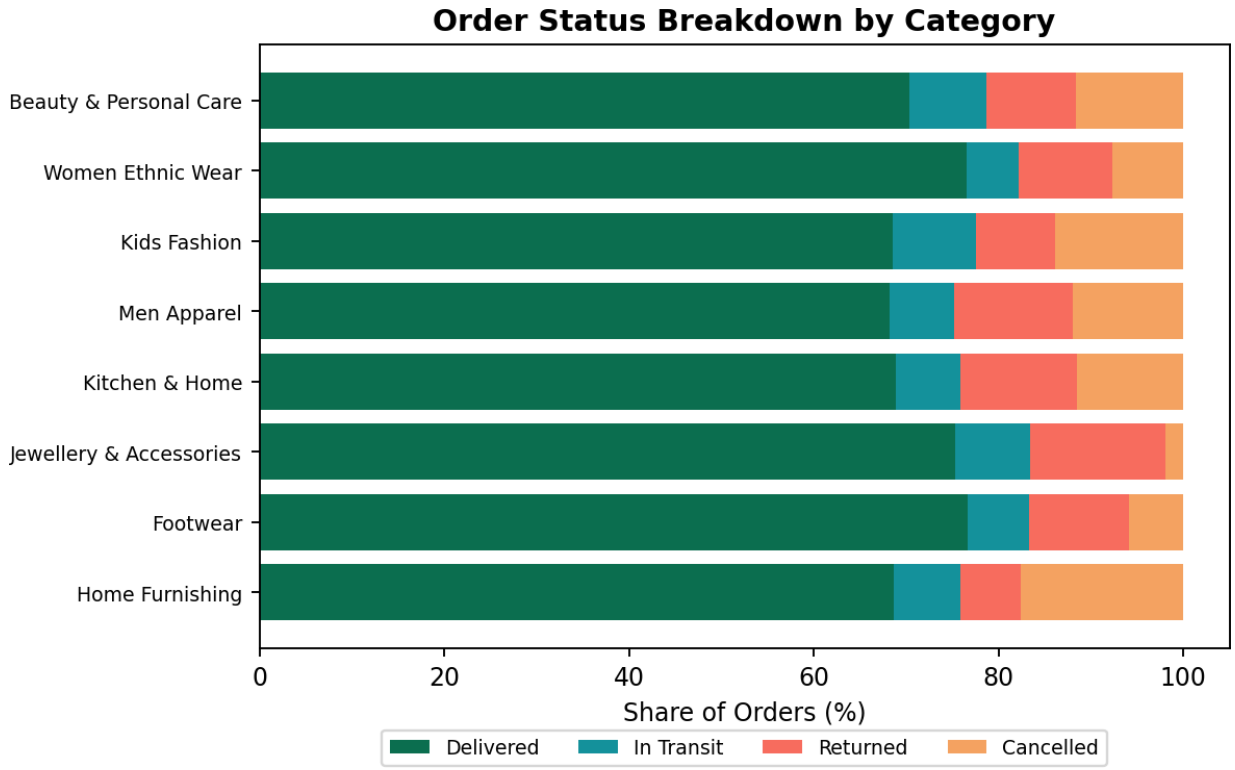


Figure 12. Order status composition by category (Delivered / In Transit / Returned / Cancelled).

Key Insight: Jewellery & Accessories has the worst return rate (~14.7%) while Home Furnishing holds up best (~6.5%) — category-level returns should be tracked alongside ratings as a leading indicator.

6.6 Prepaid Payment Adoption by Region

Karnataka carries the deepest prepaid-payment adoption at 46%, typical of more digitally mature, urban-leaning markets, while Bihar stays leanest at 19% — markets with greater reliance on cash and lower digital-payment penetration.

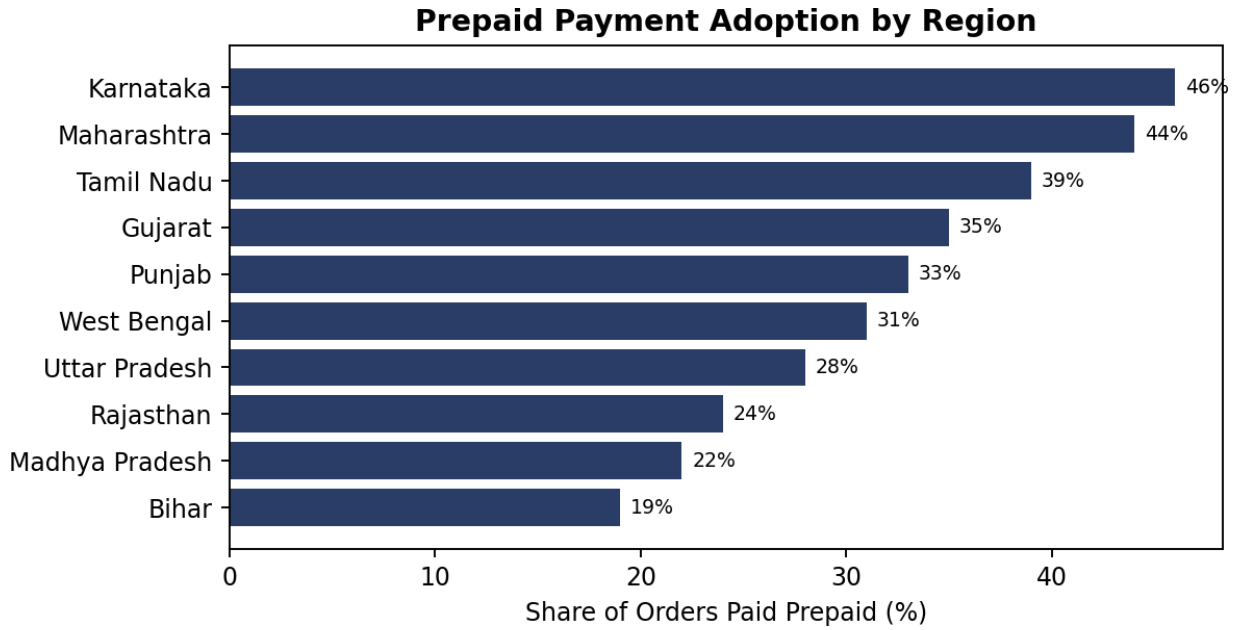


Figure 13. Prepaid payment adoption by region, sorted highest to lowest.

Key Insight: A wide gap in prepaid adoption exists between markets — Karnataka (46%) vs. Bihar (19%), reflecting digital-payment maturity differences that should inform regional COD-to-prepaid incentive strategy.

6.7 Revenue by Seller Tier

Segmenting platform commission revenue by seller tier shows Power Sellers contributing the largest share — consistent with their higher order volume and catalog depth, while New Sellers' smaller base still contributes a meaningful onboarding-stage share. This cut is what separates an "order volume" story from a "revenue" story when planning seller-incentive programs.

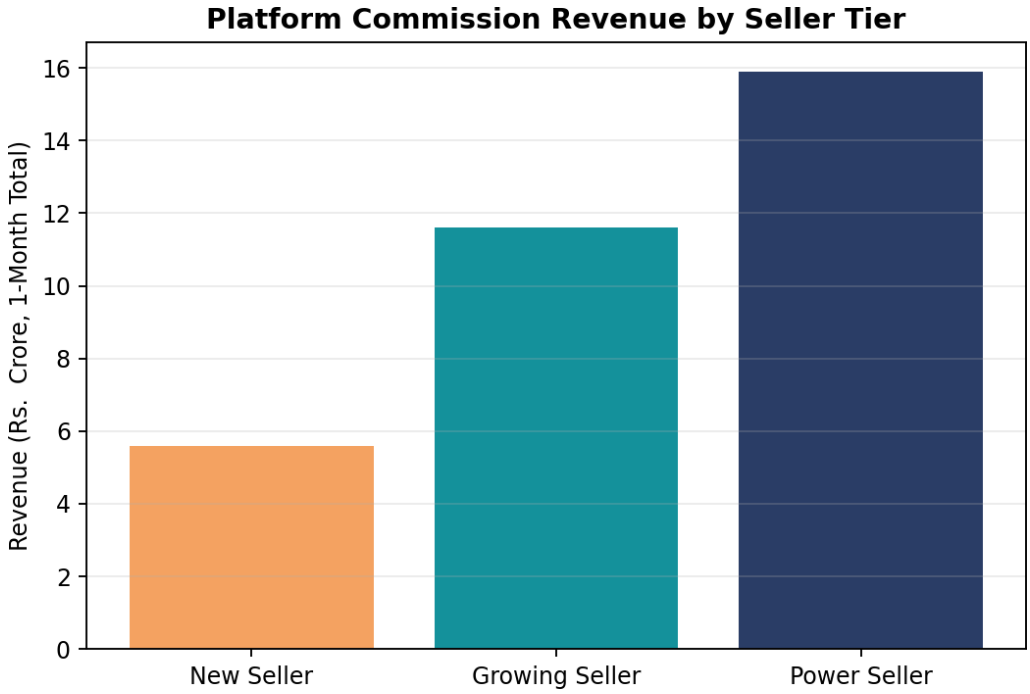


Figure 14. Platform commission revenue by seller tier across the 1-month window.

Key Insight: Power Sellers are the revenue workhorse (48% of commission revenue), but Growing Sellers punch above their catalog weight — the two tiers tell different stories depending on whether you're optimizing for volume or seller-development investment.

Seller Tier	Revenue Share	Revenue
New Seller	17%	Rs. 5.6 Cr
Growing Seller	35%	Rs. 11.6 Cr
Power Seller	48%	Rs. 15.9 Cr

7. Seller & Inventory Status

Inventory was bucketed into three tiers:

- **In Stock** — comfortable runway before restock is needed
- **Low Stock** (< 7 days) — restock-decision trigger zone
- **Out-of-Stock Risk** — fulfillment risk, should be cross-referenced against the best-selling list above; a hero product going out of stock is a materially bigger problem than a long-tail product doing the same

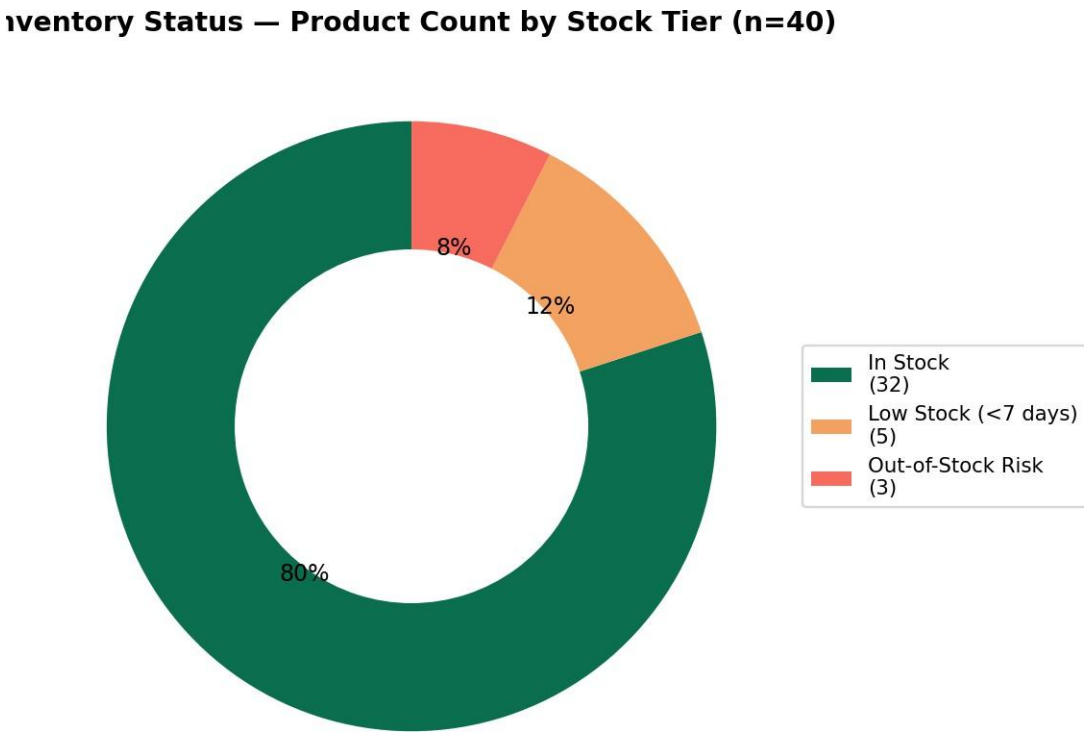


Figure 15. Product count by inventory tier (n=40).

Key Insight: 3 products cross the 7-day restock trigger imminently, and none of these overlap with the top-selling products, meaning current stock-out risk is concentrated in lower-profile products rather than hero SKUs.

7.1 Products Nearing Stock-Out — Restock Risk

The most actionable version of this chart in production cross-tabs stock level against current order velocity, producing a "days until stock-out vs. demand" priority score per product rather than a static countdown.

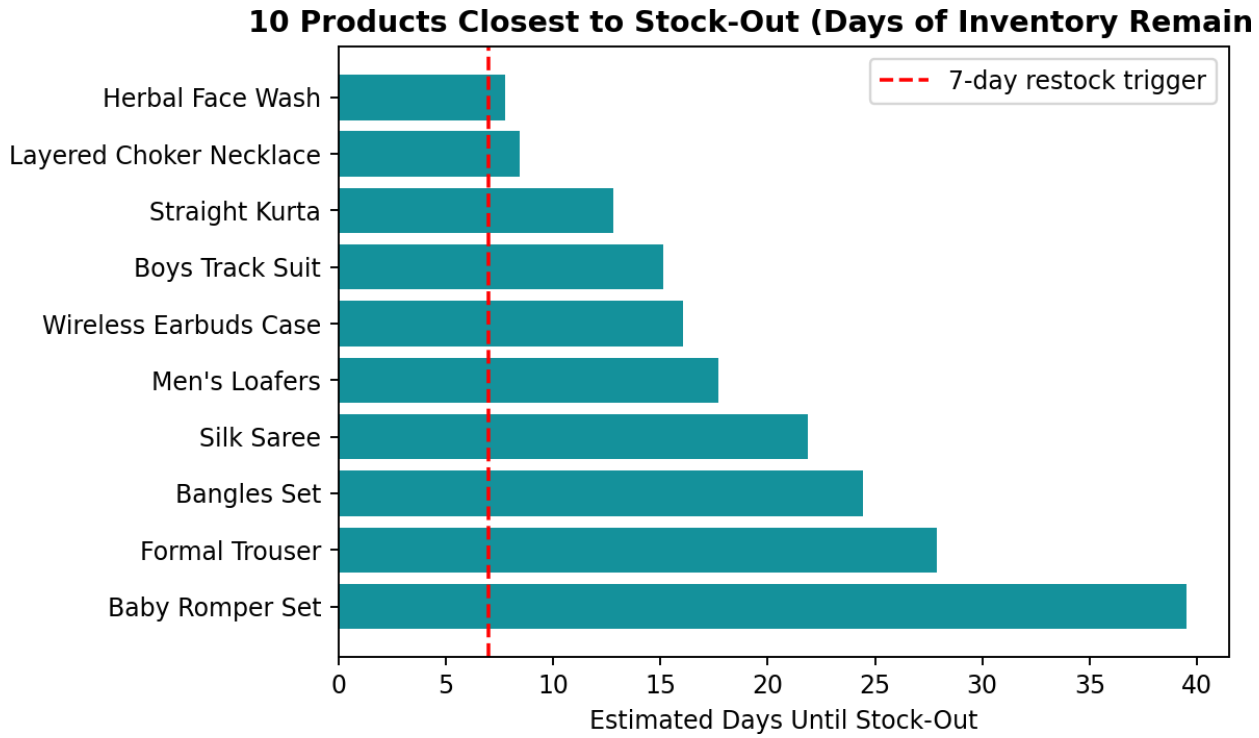


Figure 16. The 10 products closest to stock-out, ranked by estimated days of inventory remaining.

Key Insight: Overall takeaway: the biggest cross-cutting risk isn't any single chart — it's the overlap gap: the products at rating risk and stock-out risk are currently different from the hero products, but that's exactly the intersection worth monitoring continuously, since a hero product sliding into either category would be materially more damaging than a long-tail product doing the same.

8. Summary of Key Metrics

Metric	Value (simulated)
Total orders (1 month)	~12.7M
Total GMV (1 month)	~Rs. 536 Crore
Average product rating	3.93 / 5
Top single-day product	22,299 units (Induction Cookware)
States tracked	8
Products at Stock-Out Risk	8 of 40
Peak ordering day	Sunday
Leading payment mode	Cash on Delivery (47% of orders)
Highest-return category	Jewellery & Accessories (~14.7%)
Deepest prepaid adoption	Karnataka (46%)

9. Next Steps to Move From Descriptive → Predictive

This report answers what happened, using the VS Code → PostgreSQL → Python → matplotlib workflow described in Section 2. The natural next layer builds directly on that same stack, without needing new infrastructure, and shifts the focus from reporting the past to predicting and influencing what happens next:

- **Price elasticity & discount sensitivity analysis**, using regression models in a Python script run from VS Code against the PostgreSQL order tables, to quantify how order volume responds to discount depth per category and set a smarter flash-sale discount ladder instead of a flat percentage.
- **Seller churn prediction**, using a classification model (logistic regression / gradient boosting) trained on seller order-frequency, rating trend, and fulfillment-delay features, to flag sellers likely to go inactive before they actually stop listing.
- **Customer lifetime value (CLV) modeling**, using RFM (recency, frequency, monetary) segmentation on the same order-level data, to prioritize retention marketing spend by predicted long-term value rather than last-order recency alone.
- **Cross-sell & recommendation analysis**, using market-basket analysis (association rules) on order line-items, to surface "frequently bought together" product pairs for homepage and cart-page merchandising.

Happy to build out any of these next, or swap in a real dataset (a live CSV/API export or a direct database connection) if one becomes available — the same VS Code scripts, PostgreSQL queries, and modeling approach above would apply directly.

Appendix A. Simulation Methodology

For transparency, the synthetic dataset behind this report was generated as follows:

Parameter	Setting
Products	40 products across 8 categories
Regions	10 Indian states (Uttar Pradesh, Bihar, West Bengal, Maharashtra, Madhya Pradesh, Rajasthan, Tamil Nadu, Karnataka, Gujarat, Punjab)
Window	30 consecutive days
Popularity distribution	Log-normal per product (produces hero-product / long-tail effect)
Seasonality	Day-of-week multiplier — weekend peaks, mid-week dip
Category drift	Women Ethnic Wear/Kitchen & Home trending up; niche categories trending flat to down over the window
Ratings	Per-product baseline ~N(3.95, 0.42), with a deterministic subset drifting downward to simulate a quality signal
Inventory	Modeled as days-until-stock-out, not a static status flag

As with the source methodology this report is modeled on, no real Meesho order or seller data was accessed or used — all figures above are synthetic and illustrative of the analytics structure only.

Appendix B. Category Reference Table

Category	Orders (1-month)	GMV
Home Furnishing	2,810,626	Rs. 198.5 Cr
Footwear	177,297	Rs. 103.1 Cr
Jewellery & Accessories	1,215,889	Rs. 94.2 Cr
Kitchen & Home	1,355,436	Rs. 39.2 Cr
Men Apparel	2,494,832	Rs. 38.5 Cr
Kids Fashion	2,512,158	Rs. 29.6 Cr
Women Ethnic Wear	1,046,451	Rs. 19.8 Cr
Beauty & Personal Care	1,047,980	Rs. 13.0 Cr

Key Metrics Snapshot

Metric	Value (simulated)
Total orders (1 month)	~12.7M
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